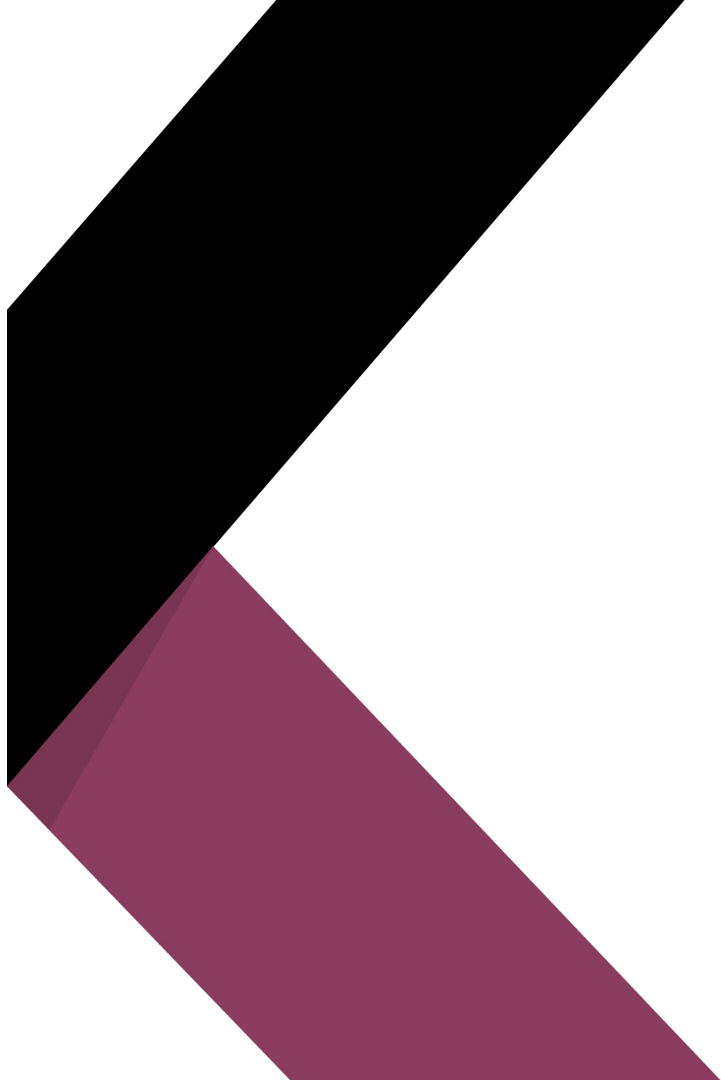


CAPL

Panel Editor





Panel Editor

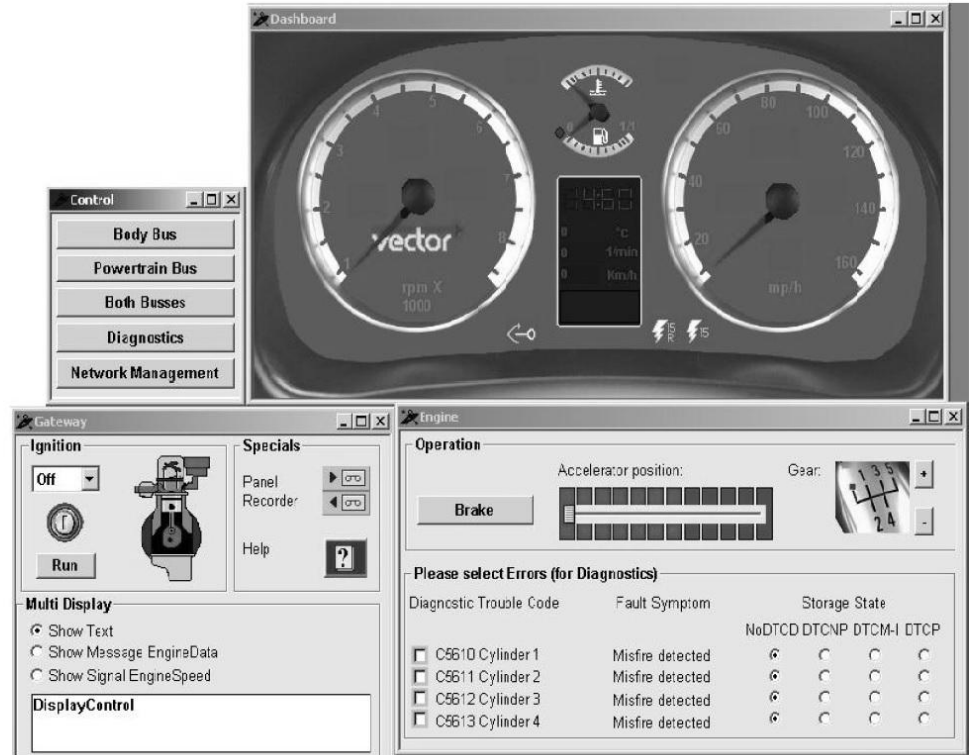


Panel Editor Introduction

A **panel** is a user- configurable graphical interface. Every control that can be placed on a panel is called an element. These elements-buttons, switches, sliders, and so on – can be placed anywhere on the panel to create the look and feel of automotive control panels.

The **Panel Editor** is the environment used for creation and arrangement of elements for displaying and controlling variables with values to be changed, and for assigning variables to all of these control and display elements.

It's also possible to put text and static bitmaps on the panel.



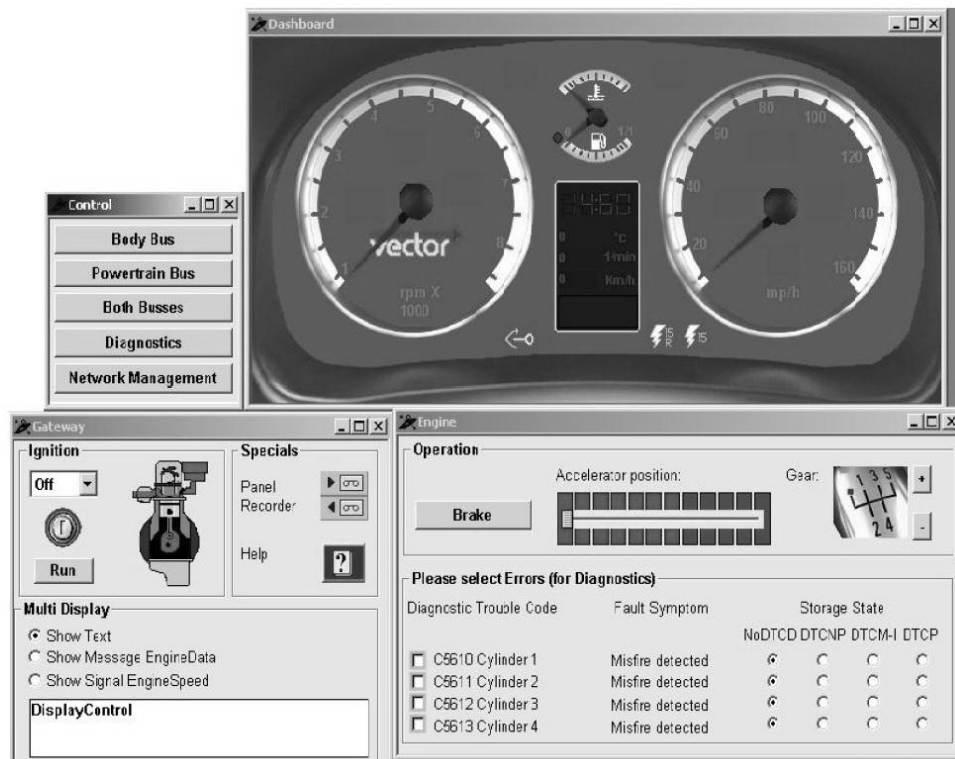


Panel Editor Introduction

It's possible to interact and change the values of these variables during measurement by manipulating the elements on the panels. The network node models react to changes in the variable values and then execute the appropriate actions (i.e. sending out a message).

Not only the user can change the values of the variables, but also CAPL programs can be used to change these values when certain events occur.

A clear advantage of using panels is that they increase your ability to interpret input and output bus data, such as on a dashboard. When a node receives a message, the node can display the data bounded by the message onto a panel.





To use the **Panel Editor** you must have CANoe.

Panel Editor is a separate application, it can be run in stand-alone mode; however it's much easier to create panels in a **Panel Editor** opened through CANoe because the variables to be associated with the elements are in the databases associated to CANoe.

The database has all the variables defined for the elements on the panel, so you will save time and effort when creating a panels because the **Panel Editor** recalls the variables defined in the database.



The variables that CANoe uses to describe external events are called environment variables and these are the same variables that can be used to associated with panel elements.

Environment variables can be used both as input and output variables of network nodes, such as sensor signals and actuator signals.

Environment variables can represent switch positions, temperature, or engine speed. Character string variables can be used to store text.



Environment variables are defined in the database by the following system parameters:

- Symbolic name of the environment variable
- Value type (i.e. Integer, String, Float, Data)
- Access rights that ECUs have when they access environment variables (i.e. Read – the assigned ECU can read the environment variable; Write – the assigned ECU can write to the environment variable; Read/Write – the assigned ECU can read from and write to environment variable)
- Unit (physical quantity of the environment variable)
- Initial value (at start of the simulation – only available for environment value type Integer and Float)
- Minimum and maximum of a valid physical value (only available for environment value type Integer and Float)
- Length of data field in bytes (only for environment variables of value type Data)



It is possible during measurement to manipulate the environment variables by manipulating the controls on the panels.

The network node will react to the changes to the environment variable and then execute the appropriate actions (i.e. sending out a message on the bus).

The new value is retrieved by the **getVlue ()** in CAPL

CAPL programs can also change the values of associated environment variables when certain event occur using the **putValue ()** function. The change will visually affect the corresponding element in the panel.



There are three methods available to start the **Panel Editor**:

- The shortcut button on the Canoe toolbar
- The menu command **Panel** □ **Configure panels**
- Select a panel in the **Configured Panels** section of the Panel configuration dialog and then click the **Edit** button. This will ensure that the database of your CANoe configuration will be automatically associated to the **Panel Editor**.

If you try to open an existing panel for modification, right-click on the title bar of the panel and select **Edit**.

If the database does not contain the environment variable names associated with the panel's indicator and control elements or if there is an element not associated to a variable yet, you will receive a warning message



To set the properties of your panel in the **Panel Editor**, click on the **Window Setting** button on the toolbar or select **Options** ☐ **Window setting**.

The first thing your panel needs is a title. This is not just to distinguish each panel from another, but helps the CANoe operator by displaying the name of the panel on the Windows taskbar at the bottom of the screen.

CAPL functions like **putValueToControl ()** also need a panel title passed as a parameter to process.

This is also the section in which you can specify the size of the panel, the fonts for each elements and the background color.



Elements allow a variety of value formats to be assigned. In addition to hexadecimal, decimal, string and bitmaps, symbolic values from CAN database (CANdb file) can be selected and/or displayed on the panel.

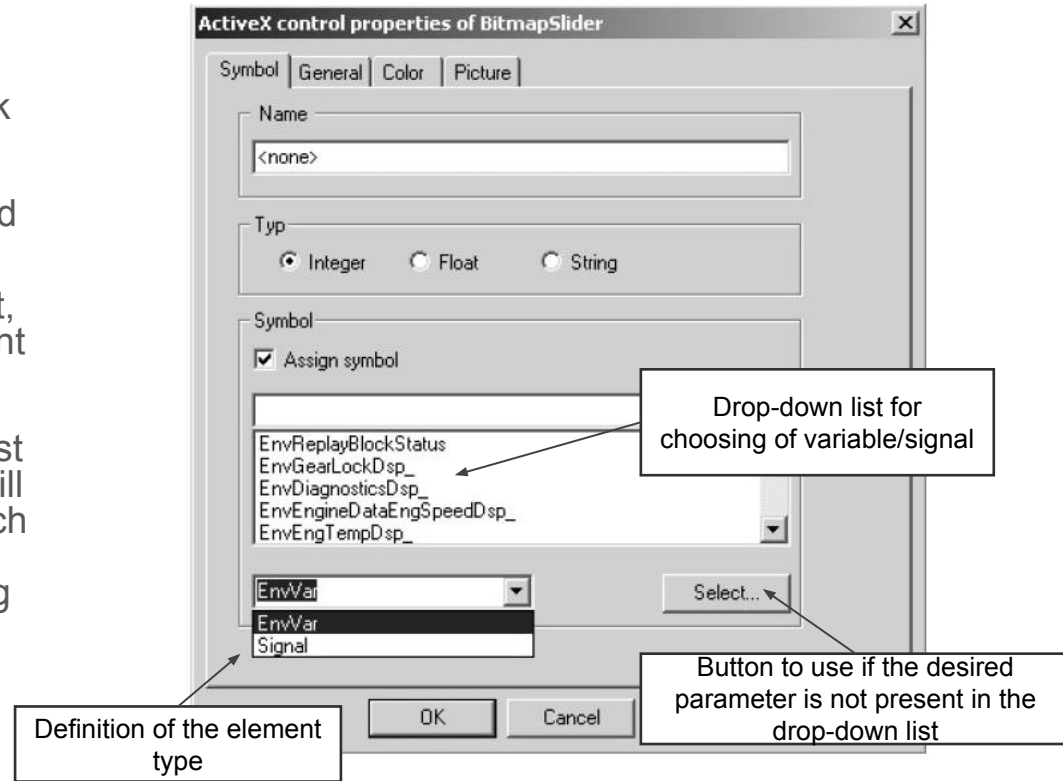
The **Panel Editor** already has a set of pre-defined elements that are ready for use. Each element has a corresponding button on the toolbar that you can click and place on the panel.

- **Control Elements:** are used to represent value from a sensor. They can let you change the values assigned to the environment variables (i.e. switch, push button)
- **Display Element:** are used to represent values from an actuator and allow let you display changes in the values assigned to environment variables (i.e. meter, multi-control)
- **Static Element:** are elements to which no environment variables can be assigned (i.e. label, bitmap)
- **I/O Controls:** they can be used either to regulate input values resulting from your programmed actions or to display the resulting output values (i.e. bitmap push button, bitmap switch, value table box etc.)



To configure a new element after dragging it onto the panel, right-click on it and select **Configure this element** or just double left-click on it. This action will open a dialog where you can specify the value type or data type and the aspect on the element (i.e. font and background colour if is possible).

Before an element can be used for input/output, it must be associated with either an environment variable or a signal defined in the database. To do this, select either **EnvVar** (environment variable) or **Signal** from the drop-down list. A list of available environment variables or signals will immediately appear in a list selection from which you can select the environment variable/signal you want. If the variables/signal you are looking for is not there, click **Select** button to find it.





An alarm state is a condition reached when a value of the panel element goes beyond its pre-defined maximum or minimum range.

Under such conditions, a visual representation of that panel element is displayed.

The dialog box is titled "Configure alarm limit values" and has a close button (X) in the top right corner. It contains two main sections: "Source of limit values" and "Limit values".

Source of limit values:

- ☐ Attributes from the database
- ☒ Min./max. values from the database
- ☐ Fixed values

Limit values:

Upper:

Lower:

At the bottom, there are three buttons: "OK", "Cancel", and "Help".



Panel Editor

Elements configuration

The **Alarm State** feature makes it possible for different panel elements to respond to a situation in which a value is above or below a previously defined value range.

It is possible to change the visual representation of the panel element when a specific value limit is reached. For example it is possible to respond to a situation in which a meter value is above or below a limit by changing colour visualization of the meter.

The 'Configure Meter' dialog box is shown with the following settings:

- Name:** <none>
- Sweep angle:**
 - from: -135 degrees
 - to: 135 degrees
- Type:** Integer (selected), Float
- Symbol:**
 - Dropdown list: motbus/NM_Engine/SleepInd, motbus/NM_Gateway_Motbus/SleepInd, motbus/GearBoxInfo/EcoMode
 - Signal: Signal
 - Select... button
- Alarm states:**
 - Display: checked
 - Configure... button
- Needle:**
 - Color: Black, Edit... button
 - Length from pivot point: 0 % of total length
 - Shape: Triangle Wide Filled
- Value range:**
 - Minimum: 0
 - Maximum: 100

Buttons at the bottom: OK, Cancel, Help.



Only bitmap (*.bmp) images can be used on panels. Bitmaps reached by right-clicking on the new element and selecting **Configure this element**.

Static display usually use a bitmap as a background. Most of the time, bitmaps are used to represent the state or condition of a variable. These bitmaps are normally used for display purposes, but can be used as inputs by clicking on the bitmap to change its state. Changing the display state, therefore, changes the value of the associated environment variable.

The elements that support I/O and bitmaps fall into two categories:

- Two-state elements
- Multi-state elements



A bitmap file for a two-state element always consists of three separate bitmap pictures. Each with the same height and width. Arranged to be horizontally adjacent to one another.

The first image appears only when no variables has been assigned to the element, the second one was used for the logical OFF state, the third one is used for logical state ON.

A bitmap file with n -states consists of $n+1$ rectangular partial bitmaps of the same height and width, arranged to be horizontal adjacent.

As for the two-state bitmap file, the first image was used when no variables has been assigned to the element, the following ones represents the $n+1$ states of the variables.



After the elements on your panel have been configured and saved, they must be associated with the desired CANoe configuration.

To associate the panels follow this steps:

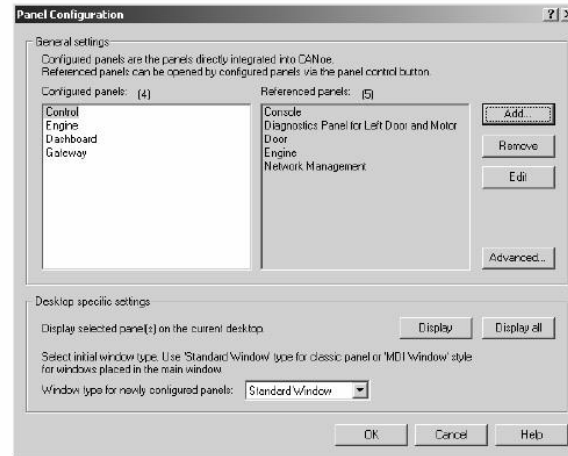
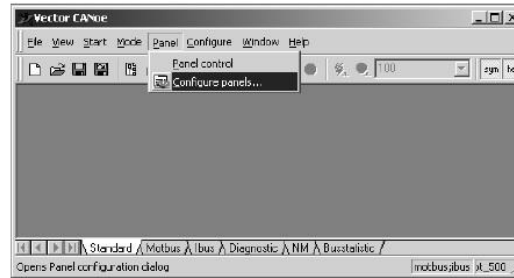
- Select the menu item **Panels** □ **Configure Panels**
- Click **Add**
- Select your panel (*.cnp)

If the panel has already been added and you made some adjustment using **Panel Editor**, you do not have to add the updated panel to CANoe again. The associated panel will automatically update once you've saved it in the **Panel Editor**.



Panel Editor

How to associate Panel to CANoe



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